3 - Relativistic Cosmological Perturbation Theory

$\rho[\vec{x},t] = \bar{\rho}[t]$ and define a new time coordinate t' such a $t' = t + \delta t[\vec{x},t]$ $\quad \delta t \ll t$

$$\rho'[\vec{x},t'] \equiv \bar{\rho}\left[t[\vec{x},t']\right]$$

when evaluate ρ' on hypersurface $t' = c^{st}$, it will in general depend on $\vec{x}$.

$$\bar{\rho}[t] = \bar{\rho}\left[t' - \delta t[\vec{x},t]\right] \simeq \bar{\rho}[t'] - \dot{\bar{\rho}}\,\delta t[\vec{x},t]$$
$$\equiv \bar{\rho}[t'] + \delta\rho[\vec{x},t'] \quad !!$$

To overcome this problem : 2 ways -

1) Calculate only metric perturbations which are gauge invariant. Bardeen variables.
2) fix the gauge and keep track of all perturbations metric and matter.

Here use (conformal) Newtonian gauge.

Restrict ourselves to scalar part. of metric. and energy densities. Ignore curvature = flat FLRW spacetime.

$$ds^2 = a^2[\eta]\left(-\left(1+\frac{2\phi}{c}\right)c^2 d\eta^2 + \left(1-\frac{2\phi}{c^2}\right)\delta_{ij}dx^i dx^j\right)$$

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g_{\mu\nu}$$

$$\delta G_{\mu\nu} = -\frac{8\pi G}{c^4}\,\delta T_{\mu\nu}$$

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$$

- $g_{\alpha\beta} = a^2[\eta] \begin{pmatrix} -(1+2\phi/c^2) & 0 \\ 0 & (1-2\phi/c^2)\delta_{ij} \end{pmatrix}$

- $g^{\alpha\beta} = \frac{1}{a^2[\eta]} \begin{pmatrix} -(1-2\phi/c^2) & 0 \\ 0 & (1+2\phi/c^2)\delta^{ij} \end{pmatrix}$

used $\frac{2\phi}{c^2} \ll 1$ so $\frac{1}{1+2\phi/c^2} \simeq 1 - 2\phi/c^2$

$\longrightarrow$. $G_{00} \equiv R_{00} - \frac{1}{2} g_{00} R$

$$= -3\frac{\mathcal{H}^2}{c^2} - 2\frac{\nabla^2\phi}{c^2} + 6\mathcal{H}\frac{\phi'}{c^4}$$

. $G_{0i} = -2\mathcal{H}\frac{\partial_i\phi}{c^3} - 2\frac{\partial_i\phi'}{c^3}$

. $G_{ij} = \left(2\frac{\mathcal{H}'}{c^2} + \frac{\mathcal{H}^2}{c^2} - 8\mathcal{H}'\frac{\phi}{c^4} - 6\mathcal{H}\frac{\phi'}{c^4} - \frac{2\phi''}{c^4} - 4\frac{\mathcal{H}^2\phi}{c^4}\right)\delta_{ij}$

$$T^\mu_{\ \nu} = \bar{T}^\mu_{\ \nu} + \delta T^\mu_{\ \nu}$$

$$= \left(\bar{e} + \delta e + (\bar{P} + \delta P)/c^2\right)\bar{U}^\mu \bar{U}_\nu$$

$$+ \left(\bar{e} + \bar{P}/c^2\right)\left(\delta U^\mu \bar{U}_\nu + \bar{U}^\mu \delta U_\nu\right)$$

$$+ (\bar{P} + \delta P)\,\delta^\mu_{\ \nu}$$

$$T^0_0 = -(\bar{\rho} + \delta\rho)c^2$$

$$T^i_0 = -(\bar{\rho} + \bar{P}/c^2)cv^i$$

$$T^0_j = (\bar{\rho} + \bar{P}/c^2)cv_j$$

$$T^i_j = (\bar{P} + \delta P)\delta^i_j$$

if define momentum density as $q^i \equiv (\bar{\rho} + \frac{\bar{P}}{c^2})v^i$

$$T_{\mu\nu} = \sum_X T_{\mu\nu,X}$$

X = photons, baryons, dark matter ...

then $\delta\rho = \sum_X \delta\rho_X$; $\delta P = \sum_X \delta P_X$

and $q^i = \sum_X q^i_X$.

$$G_{00} = -\frac{8\pi G}{c^4} g_{0\mu} T^\mu_0$$

$$-3\frac{\mathcal{H}^2}{c^2} - 2\frac{\nabla^2\phi}{c^2} + 6\mathcal{H}\frac{\phi'}{c^4} = -\frac{8\pi G a^2}{c^2}\bar{\rho}\left(1 + \delta + 2\frac{\phi}{c^2}\right)$$

$$\delta \equiv \frac{\delta\rho}{\bar{\rho}}$$

$$\mathcal{H}^2 = \frac{8\pi G a^2}{3}\bar{\rho}$$

$$\boxed{\nabla^2\phi = 4\pi G a^2 \bar{\rho}\delta + 3\frac{\mathcal{H}}{c^2}(\mathcal{H}\phi + \phi')}$$

$\phi' \sim \mathcal{H}\phi$ and recall $\mathcal{H} \sim 1/\eta$

Fourier transform: $\nabla^2 \to \frac{1}{k^2}$

for $kc\eta \gg 1$ GR correction term is negligible

$$G_{ij} = -\frac{8\pi G}{c^4} g_{ij} T^{\mu}{}_{j}$$

$$\to \mathcal{H}' = -\frac{4\pi G a^2}{3}\left(\bar{\rho} + 3\frac{\bar{P}}{c^2}\right)$$

$$\to \boxed{\phi'' + 3\mathcal{H}\phi' + (2\mathcal{H}' + \mathcal{H}^2)\phi = 4\pi G a^2 c_s^2 \bar{\rho}\delta}$$

$$\nabla_\mu T^{\mu}{}_{\nu} = 0$$

$$\to \partial_\mu T^{\mu}{}_{\nu} + \Gamma^{\mu}_{\mu\alpha} T^{\alpha}_{\nu} - \Gamma^{\alpha}_{\mu\nu} T^{\mu}_{\alpha} = 0$$

Consider $\nu = 0$ component:

$$\partial_0 T^0{}_0 + \partial_i T^i{}_0 + \Gamma^i_{i0} T^0{}_0 + \Gamma^j_{ji} T^i{}_0 - \Gamma^i_{00} T_i{}^0 - \Gamma^i_{0}{}^{j} T^i{}_j = 0$$

$$\bar{\rho}' + \bar{\rho}'\delta + \bar{\rho}\delta' + (\bar{\rho} + \bar{P}/c^2)\frac{\partial \vec{v}}{\partial x} + 3(\mathcal{H} - \phi'/c^2)(\bar{\rho} + \bar{P}/c^2) + 3\mathcal{H}(\delta\rho + \delta P/c^2) = 0$$

. 0th order : $\bar{\rho}' + 3\mathcal{H}\bar{\rho}\left(1 + \frac{\bar{P}}{\bar{\rho}c^2}\right) = 0$

. 1st order :

$$\boxed{\delta' + \left(1 + \frac{\bar{P}}{\bar{\rho}c^2}\right)\left(\frac{\partial \vec{v}}{\partial \vec{x}} - 3\frac{\phi'}{c^2}\right) + 3\mathcal{H}\left(\frac{c_s^2}{c^2} - \frac{\bar{P}}{\bar{\rho}c^2}\right)\delta = 0}$$

Euler come from the $\nu = i$ component.

$$\boxed{\vec{v}' + \mathcal{H}\vec{v} - 3\mathcal{H}\frac{c_s^2}{c^2}\vec{v} = -\frac{\partial \phi}{\partial \vec{x}} - \frac{c_s^2}{1 + \bar{P}/\bar{\rho}c^2}\frac{\partial \delta}{\partial \vec{x}}}$$

Super cluster of galaxies. Laniakea or Saraswati have sizes $\sim$ 150 Mpc

$R_h(z=0) \sim 14.4$ Gpc